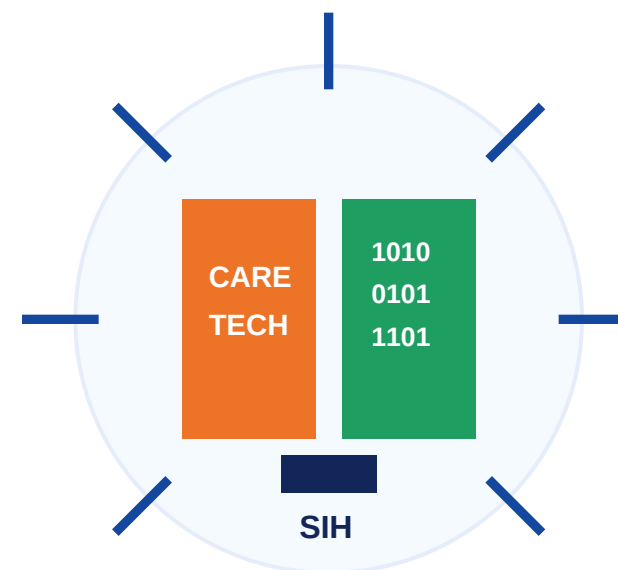


TITLE PAGE

- • **Problem Statement ID** – 67
- • **Problem Statement Title** – Accessibility and quality of public healthcare services, particularly in rural and underserved areas.
- • **Theme** – MedTech
- • **PS Category** – Software / MedTech
- • **Team Name** – CareMatrix
- • **Team ID** – CareMatrix-67



CareMatrix — AI-Powered Offline-First Smart Clinic Platform for Rural Healthcare

Proposed Solution

> **Unified Frontline Healthcare Ecosystem:**

A synchronized digital clinic suite connecting visiting Doctors, Primary Health Centres (PHCs/CHCs), field Nurses/ASHA workers, and rural patients into one single workflow.

> **Multimodal Intelligent Intake:**

Seamlessly digitizes paper prescriptions, lab reports, manual vitals, and multilingual voice dictations into standardized structured medical records in seconds.

> **Resilient Offline-First Edge Operation:**

Built with local browser-level OCR and indexed transactional storage, allowing uninterrupted patient consultations and intake even during complete cellular network blackout.

> **Vernacular & Voice-Enabled Empowerment:**

Overcomes rural literacy barriers through automated audio explanations, local Indian languages translation, and visual dosage schedules for safe medication adherence.

> **Zero-Infrastructure Portable QR Records:**

Generates dynamic, encrypted QR health cards that store key clinical history directly with the patient, enabling instant emergency access at any referral hospital.

Frontend & Edge Tier

React, TypeScript, Vite, Tailwind CSS with responsive progressive web app (PWA) architecture.

Offline Edge OCR

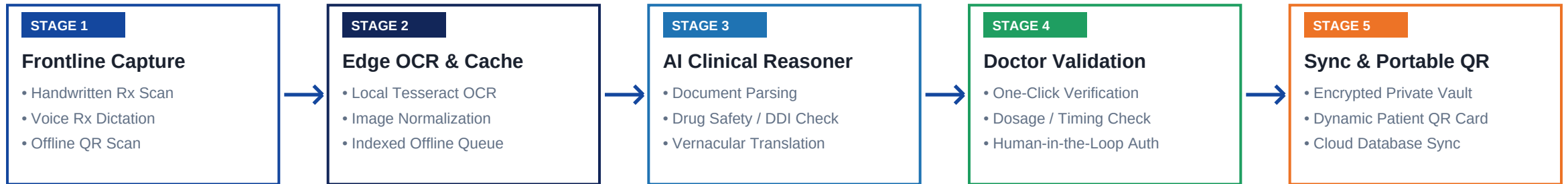
Client-side Tesseract.js engine with localized transcription queues for zero-internet data capture.

Clinical AI Reasoning

Multimodal document parsing, Voice Rx structuring and automated CDSCO/WHO drug safety checks.

Secure Cloud & Vault

Firebase Firestore with Role-Based Access Control and AES-password-guarded Private Vault.

SYSTEM ARCHITECTURE & CLINICAL WORKFLOW FLOW DIAGRAM

Input Modes: Camera scan, image upload, portable QR scan, Voice Rx dictation, manual vitals entry.

Workflow: Capture -> Extract -> Validate -> AI Drug Safety Check -> Store Locally/Cloud -> Doctor Review.

READY

Working Prototype Validated

The working solution already features core operational workflows: digital prescription scanning, voice Rx intake, longitudinal vitals tracking, dynamic QR access, appointment booking, and role-based access control.

LOW COST

Zero Hardware Overhead & Universal Compatibility

Powered entirely by lightweight browser technologies. Runs smoothly on budget Android tablets, standard smartphones, and village clinic laptops without requiring expensive servers or dedicated IT infrastructure.

OFFLINE-FIRST

Engineered for Low & Intermittent Connectivity

Offline-first architecture empowers primary health centres (PHCs), mobile medical vans, and tribal health outposts to record data locally during outages and automatically sync once network is restored.

SAFETY FIRST

Strict Clinician Decision Support & Safety Guardrails

AI functions exclusively as Clinical Decision Support (CDS) to flag drug-drug interactions and assist transcription. Registered doctors always retain absolute review and sign-off authority.

SCALE

Scalable Public Health Rollout & Low TCO

Cloud-native database model allows rapid multi-district deployment with minimal operational cost, fitting seamlessly within national digital health mission frameworks.

> Rural Patients & Families:

Eliminates lost paper prescriptions, provides audio-visual dosage instructions in local languages, prevents medication mistakes, and guarantees lifetime portable access to medical history.

> Frontline Doctors & Medical Officers:

Reduces manual consultation paperwork by up to 70%, flags potential adverse drug-drug interactions automatically, and provides instant visibility into past vitals and prior lab findings.

> Nurses, ASHA Workers & Clinic Staff:

Streamlines health camp vitals intake, lab report scanning, and automated patient follow-ups, reducing administrative stress and speeding up patient queue clearance.

> Public Health Administration & Government:

Provides real-time district-level visibility into disease trends, epidemic outbreaks, pharmaceutical stock demands, and clinic visit volumes with zero manual reporting overhead.

> Continuity of Care & Referral Integrity:

Dynamic QR health cards ensure seamless, error-free patient handoffs between rural sub-centres (PHCs) and district tertiary hospitals without duplicate record creation.

> **Ayushman Bharat Digital Mission (ABDM) - National Health Authority (NHA), Govt. of India**

Frameworks and data standards for longitudinal Electronic Health Records (EHR), unified ABHA patient identifiers, and health facility registries across primary and secondary tiers.

> **WHO Model List of Essential Medicines & CDSCO Prescribing Safety Guidelines**

Standardized pharmacology datasets, contraindication taxonomies, and multi-drug interaction screening rules designed to protect patients from polypharmacy risks in primary care.

> **Peer-Reviewed Clinical Natural Language Processing & Medical Informatics Research**

Grounded in state-of-the-art multimodal vision-language architectures for robust document entity extraction, medical handwriting parsing, and clinical dialogue summarization.

> **Tesseract OCR Engine & Progressive Web App (PWA) Offline Synchronization Protocols**

Validated client-side edge computing techniques enabling zero-infrastructure optical character recognition, local indexed data caching, and fault-tolerant cloud synchronisation.

> **ISO/IEC 18004 Standards for High-Capacity 2D QR Matrix Data Interchange**

Design and cryptographic safeguarding of portable two-dimensional matrix cards for offline patient record continuity across remote health clinics and district hospitals.